

Attention-Sink Signatures Dissociate During Vision–Language Pretraining

Anonymous author(s)

Abstract. The attention sink is described as one phenomenon and measured as three: attention concentration on the sink position (Sink_1^e , per query head), a near-zero value-norm there (v-ratio, per KV head), and an outsized residual-stream norm there (h-ratio, per layer, which we read as a massive-activation proxy). In text language models the three co-occur reliably enough that a study can measure one of them and generalise to the sink. That inference fails if they come apart. Whether a lever removed the sink then depends on which signature someone happened to track.

We watch the three form, in multimodal pretraining with a randomly initialized decoder. The model is a 222M-parameter vision–language model, a SigLIP-B/16 encoder feeding a randomly initialized SmoLLM2-135M-architecture decoder whose position 0 is the first image token, with no BOS. Four levers: softmax attention, output-gated softmax, unnormalized sigmoid attention, and initialization from a pretrained text LM.

They come apart. Across $n = 2\text{--}3$ seeds per arm the levers reach four distinct corners of the three-signature space, and no two arms share a triple. The value-norm axis alone is strongly drained, mildly drained, or amplified, and which one depends on the lever. On a low-repetition run over a billion fresh tokens (2.39 effective visual epochs) the massive-activation proxy more than doubles while concentration stays at exactly zero. The per-head relationship between concentration and value-norm flips sign from arm to arm. Prior text-only work separates massive activations from attention sinks. We track all three together, under randomly initialized decoders in a multimodal model, and find that each one, value-norm drain included, answers separately to ordinary training choices.

1. Introduction

Decoder-only transformers pick up a habit early in training. They send a large share of attention to the first token or tokens of a sequence, whatever those tokens contain. The habit carries practical weight: streaming inference methods depend on it [1], the extreme activation outliers that ride along with it make quantization harder [2], and a survey now organizes a subfield around interpreting and removing it [3].

The sink does not arrive alone. Three measurements move together in text language models. Attention concentrates on the sink token. The value vector there drops to a near-zero norm, called value-state drain [4]. The residual stream there grows an abnormally large norm, called a massive activation [2, 5]. Gu et al. [6] report that these effects settle on the same few tokens, and Queipo-de-Llano, Arroyo et al. [7] that they appear near step 1000. The field has taken the co-occurrence as licence to treat all three as facets of one attention-sink phenomenon.

Whether they really are one phenomenon is contested. One line argues for causal unity: massive activations mathematically require representational compression, and ablating them removes both compression valleys and sink formation [7]. A related view holds that outlier-driven rescaling by attention and residual sinks is *essential* to stable training [8]. Two recent studies pull the other way, each separating a pair of the signatures by intervention. A change of normalization scheme crushes the massive-activation spike while the sink ratio survives [9]. A value-scale intervention keeps the sink while suppressing massive activations [10]. Both are text-only, and each separates at most two of the three.

Which signature a study measures therefore decides what it can conclude. If the three move independently, a lever that clears concentration while leaving the value path alone reads as a fix under one metric and a failure under the next, and two papers can disagree about whether the sink was removed while both are right about what they measured. The stakes reach past bookkeeping. Sink-

like attention has been tied to hallucination and weak visual grounding in deployed vision-language models [13–16], so which signature a mitigation actually moves has practical weight. We measure all three at once, throughout training, and test no downstream behavior here.

Vision-language models make a sharp instrument for the question. The multimodal setting changes what position 0 is: our sequence starts with 49 image tokens and has no BOS token, so the candidate sink token is a visual token, without the special-token machinery that text-LM sink accounts lean on. The multimodal sink studies we know analyze mostly frozen, already-trained backbones at inference time [11, 12], and where one tracks alignment checkpoints [11] it follows a single magnitude rather than the three signatures (§4). A reader still cannot see there whether the signatures arrive together, in sequence, or independently while they form in a decoder that starts from random weights. That takes pretraining with a randomly initialized decoder and all three logged separately from step 0.

We work at deliberately small scale. The model is a 222M-parameter nanoVLM [17], where a pre-trained SigLIP-B/16 encoder [18] feeds a randomly initialized decoder with the SmolLM2-135M architecture [19]. Four levers each target one sink-relevant mechanism (§2): standard softmax attention (*baseline*), a Qiu-style G1 output gate in our zero-initialized variant [20] (*gate*), unnormalized sigmoid attention, which removes the normalization sinks are argued to come from [6] (*sigmoid*), and initialization from the pretrained SmolLM2 text model, which imports whatever sink structure text pretraining built (*textinit*). A validated probe logs all three signatures every 100 steps. The four-arm comparison reuses a small image pool, so we re-test the central negative result under low repetition, on a fresh stream of one billion tokens (*RF*, 2.39 effective visual epochs).

Contributions.

1. **Four levers, four corners (n = 2–3 seeds/arm).** The four arms reach four different corners of the three-signature space, and no two share a triple. The value-norm ratio alone is strongly drained, mildly drained, or amplified, and which one depends on the lever (Fig. 1, Table 1). The arms are not a factorial design, so we report distinct intervention-associated profiles, not isolated causal effects.
2. **Low-repetition decoupling at 1B tokens (n = 1).** On a fresh stream at 2.39 effective visual epochs, with a held-out loss that never turns upward, the massive-activation proxy rises from an h-ratio of 1.43 to 3.22, about 2.3 $\times$. Concentration stays at exactly zero for the entire single-seed run, and no head ever crosses the sink threshold (§3.2, §5).
3. **No consistent-sign head-level relationship.** The per-head correlation between concentration and value-norm flips sign across arms (+0.76 baseline $\rightarrow$ -0.79 textinit, pooled -0.20, over 90 KV groups per arm). We report these descriptively (§3.3).

Text-only work already separates massive activations from concentration [9, 10], so decoupling itself is not new and we do not claim it. The new part is the conjunction: dense, joint tracking of all three signatures under randomly initialized decoders in a multimodal model, with value-norm drain as a third axis that moves on its own.

2. Setup

Model and token layout. All runs use a 222M-parameter nanoVLM [17]. A pretrained, trainable SigLIP-B/16 vision encoder [18] feeds a decoder with the SmolLM2-135M architecture [19] through a learned modality projector. The decoder has 30 layers of grouped-query attention, 9 query heads per layer sharing 3 KV heads, so it has 270 (layer, query-head) pairs but only 90 (layer, KV-group) value projections, a distinction that matters in §3.3. It trains **from random initialization** in every arm except *textinit*, because the point is to watch the signatures form. Each sequence holds 49 image tokens as a causal prefix, then 79 left-padded text tokens, 128 in all. **Position 0 is the first image token, and there is no BOS token**, so what happens there is a property of the visual prefix rather than inherited BOS machinery. *textinit* is the designed exception: it imports first-position structure from text pretraining and shows a sink before any multimodal step (§3.1).

Arms. We use four training levers. Each lever targets one sink-relevant mechanism. Everything else stays byte-identical across arms.

arm	attention	LM init	ViT init	lever precedent
<i>baseline</i>	softmax	random	pretrained	—
<i>glgate</i>	softmax + elementwise σ -gate (zero-init, post-SDPA)	random	pretrained	G1 gating [20]
<i>sigmoid</i>	unnormalized sigmoid, no softmax	random	pretrained	Gu et al. [6]
<i>textinit</i>	softmax	pretrained SmolLM2-135M	pretrained	— (novel control)

The *glgate* and *sigmoid* levers have established sink effects in text-only models [20, 6]. *textinit* has no precedent in that literature and works as an inheritance control, carrying in whatever sink structure text pretraining already built into SmolLM2.

A scale confound in our gate variant. Qiu et al. [20] use ordinary initialization for the G1 gate. We initialize at exactly zero, so the sigmoid opens at $\sigma(0) = 0.5$ and the arm begins as a half-scale attention-output intervention as well as a gating one. We call it **Qiu-style G1 in our zero-initialized variant** throughout (§5).

Data and the two training regimes. The four-arm comparison trains on four curated subsets of the *cauldron* [21], about 146K images, matched at about 100M tokens per arm. *textinit* stops at 60M tokens, where its signatures have plateaued (§3.1). Reuse of that pool gives high visual-epoch counts, so a reader could treat a “no sink emerges” result as an overfitting artifact. The **RF** arm (random-fresh) answers that objection: the *baseline* recipe re-trained on a fresh FineVision stream [22] to 1B tokens, over about 4.6M natural images, at **2.39 effective visual epochs**. RF is therefore a **low-repetition** run, not a repetition-free one, since examples do repeat about 2.4 times on average. We estimate the overlap between the fresh pool and the repeated subsets at under 3%, from the config-level composition of the two pools rather than image-level deduplication. Swapping datasets trades the repetition confound for a domain-shift confound, which §5 takes up.

Three signatures, tracked separately. We log the three sink symptoms that the text-LM literature reports together, each at its own granularity, following Gu et al. [6] at fixed sequence length. The decoder has $L = 30$ layers, $H = 9$ query heads per layer, $G = 3$ KV heads per layer. Every quantity below is averaged over valid query positions and over the fixed probe batch.

Concentration ($\text{Sink}^\varepsilon_1$) is the fraction of the $L \cdot H = 270$ (layer, query-head) pairs whose mean attention to position 0 exceeds ε . We use the $\varepsilon = 0.3$ default of [6] and check $\varepsilon \in \{0.2, 0.4\}$. Cross-arm tables report the stricter $\varepsilon = 0.2$, which makes an absence claim harder to pass.

Value-norm ratio (v-ratio) is the value norm at position 0 divided by the mean value norm over the other valid positions, computed per layer and then averaged over the 30 layers. Below 1 is value-drain [4]. Above 1 is amplification. Under grouped-query attention a value vector belongs to a KV group and repeats across 3 query heads, so the ratio rests on $L \cdot G = 90$ independent value projections, not 270. This matters for §3.3.

Residual-norm ratio (h-ratio) is the residual-stream norm at position 0 divided by the mean norm over the other positions, again per layer and then averaged over the 30 layers. We call it a **massive-activation proxy**, because massive activations are normally defined by channel-level outliers [2, 5], which we never measured (§5).

All three metrics **anchor on position 0 by construction**. We state this as a measurement choice and check it: at seed 0, per-position attention mass makes position 0 the maximum-mass token in every arm (Appendix C). §5 handles the remaining seed-level caveat. The *sigmoid* arm reports the row-normalized attention view, which keeps concentration comparable across arms, and we also log the raw sigmoid mass (Appendix D).

Why a sink is expected at all. Softmax forces every attention row to sum to one, so a head with nothing worth retrieving must still put its mass somewhere. Gu et al. [6] argue the sink token absorbs that surplus and acts “more like key biases.” Our *sigmoid* arm, which removes normalization, tests that constraint directly.

Probe. The function `probe_sinks()` re-walks the decoder from the live module weights in eager mode, in fp32, with no gradients, independent of the training path, which uses fused SDPA kernels, autocast and `torch.compile`. Every call validates its hidden states against the real forward pass to a relative error below 10^{-2} , so the probe cannot drift from what the trained model computes. The probe batch is fixed across all runs and seeds, so every number here is comparable run to run, and probes fire every 100 optimizer steps, dense enough to timestamp each signature’s first threshold crossing (§3.4). Appendix F gives the probe-batch composition, token accounting and validation protocol.

Recipe. We use AdamW with weight decay 0.1, following [6], and a gradient clip of 1.0. The schedule is cosine with 3% warmup. The learning rates are 4×10^{-4} for the language model, 2×10^{-3} for the projector, and 10^{-4} for the vision encoder. We use bfloat16 autocast, `torch.compile`, and a batch size of 128. Arms in a comparison differ only in the lever under test. We use two seeds for *baseline* and *sigmoid*, three for *g1gate* and *textinit*, and one for *RF*.

Validation losses, and what they do and do not license. Training stays healthy in all reported runs. At the matched 100M-token checkpoint the held-out losses are 1.182 for *baseline*, 1.133 for *g1gate*, 1.206 for *sigmoid* and 0.877 for *textinit*, and RF reaches 0.638 at 1B tokens (Appendix F). Two cautions. *textinit* starts from a pretrained text decoder, so its lower loss reflects unequal competence rather than a lever effect, and only *baseline*, *g1gate* and *sigmoid* are equal-token, equal-initialization comparisons. The repeated-data arms also show a large train–validation asymmetry (`val_seen` near 0.44 against `val_unseen` near 1.18), the overfitting signal that motivates RF. RF has no distinct seen split, so the weaker statement its data supports is that its held-out loss falls throughout and never turns upward (§5). **We did not run MMStar or any other downstream benchmark on any arm**, so this paper makes no capability claim (§5).

3. Results

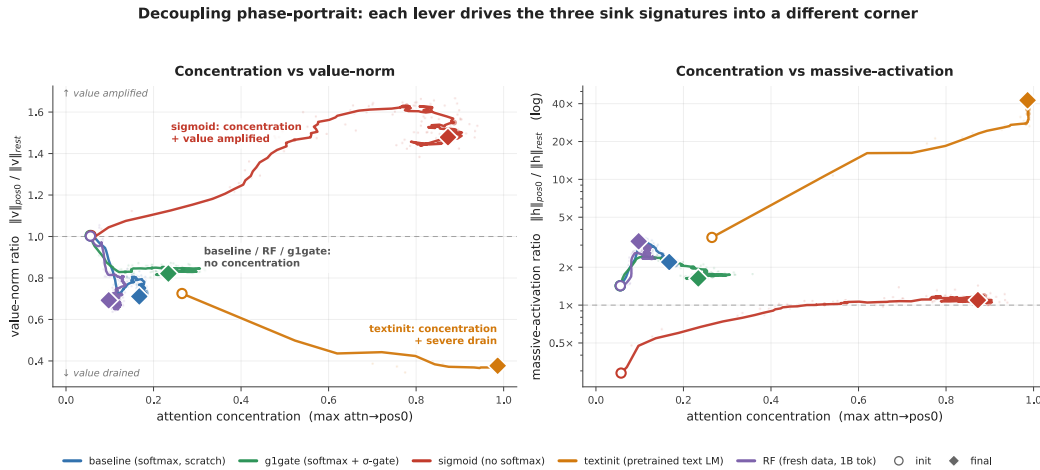


Figure 1: Decoupling phase-portrait. Trajectories of every arm in concentration-against-value-norm space (left) and against the residual-norm ratio (right, log scale). Circles mark initialization, diamonds the final checkpoint: about 100M tokens for *baseline*, *g1gate* and *sigmoid*, 60M for *textinit*, 1B for *RF*. Seed 0 unless labelled. **The horizontal axis differs from the tables:** it is the *maximum* attention→pos0 over heads, where Table 1 reports Sink_1^E , the *fraction of heads above a threshold*, so an arm can sit off zero here with $\text{Sink}_1^E = 0.000$. Paths are smoothed, end markers use raw values (Appendix D). Were the three signatures one phenomenon, these trajectories would move along one direction.

3.1 Four training levers reach four different signature corners

Table 1 collapses the seeds into per-arm ranges, each arm read at its final matched checkpoint: about 100M tokens for baseline, glgate and sigmoid, and 60M for *textinit*, whose signatures have plateaued by then. Appendix B gives the per-seed table. **No two arms share a signature triple.**

Table 1 — the four corners ($n = 2\text{--}3$ seeds per arm). Concentration is $\text{Sink}^{0.2}_1$, the fraction of the 270 (layer, query-head) pairs whose mean attention $\rightarrow$ pos0 exceeds 0.2. The v-ratio rests on 90 (layer, KV-head) value projections and the h-ratio on 30 layers (§2).

arm	concentration	value-norm	massive-activation proxy
baseline	absent (0.000)	mild drain (0.69–0.72)	moderate (1.7–2.2)
glgate	near-absent (0.004–0.011)	milder drain (0.81–0.85)	moderate (1.7–2.2)
sigmoid	strong (0.76–0.83)	amplified (1.48–1.60)	no strong asymmetry (1.1–1.3)
textinit	strong (0.56–0.85)	strong drain (0.38–0.63)	extreme (5.5–42.5)

The value-norm axis alone takes three qualitatively different directions: drained hard, drained mildly, or amplified above 1. No arm leaves it unchanged.

The corners separate pairwise on single axes. *glgate* differs from *baseline* on the value-norm axis alone, since concentration is absent or near-absent and the residual-norm ratio moderate in both, while the gate makes the baseline’s mild drain milder still, a 15–19% drain rather than none. *sigmoid* and *textinit* both reach strong concentration and then part on the *direction* of the value-norm move and on the residual-norm ratio. Each arm shows a distinct intervention-associated profile. The arms are not a factorial design, so we do not attribute these profiles to isolated causal effects of single levers. The trajectories in Figure 1 separate early and do not share one origin: *textinit* starts from a pretrained text decoder, with an inherited sink already in place.

Reproducibility differs by arm. *glgate* is tightest, at $\text{Sink}^{0.2}_1$ of 0.004, 0.011 and 0.0037 across three seeds. The corner of *textinit* repeats across seeds and its magnitudes do not: seed 0 sits far above the other two on every signature at once, at h-ratio 42.5 against 5.5–12.2, partly because at seeds 1 and 2 its peak signatures move off position 0, where our metrics anchor (§3.4, Appendix B). We therefore report the massive-activation proxy of *textinit* as a **range (5.5–42.5×) with a median near 12×** throughout, and treat the corner as the reproducible claim rather than any magnitude.

3.2 The massive-activation proxy grows across 1B low-repetition tokens while concentration stays at zero

The four-arm comparison reuses a 146K-image pool, so a reader could read its “concentration never emerges” result as an overfitting artifact. The RF arm addresses that: the identical *baseline* recipe on a fresh FineVision stream to one billion tokens, at **2.39 effective visual epochs**. Examples still repeat, about 2.4 times on average, so this is a **low-repetition** run rather than a repetition-free one. Its held-out loss falls through the full billion tokens and never turns upward (§2, §5).

Concentration never arrives. $\text{Sink}^{0.3}_1 = 0.000$ across the entire 0 $\rightarrow$ 1B run, with no head of the 270 crossing the threshold at any of about 700 probes. The massive-activation proxy, in the same run, keeps growing far past warmup.

Table 2 — RF (fresh stream, 2.39 visual epochs, $n = 1$ seed), init $\rightarrow$ 1B tokens.

signature	@ init	@ 1B	net
$\text{Sink}^{0.3}_1$ (concentration)	0.000	0.000	flat zero, entire run
max attn $\rightarrow$ pos0	0.056	0.098	$\approx$ flat, far below threshold
h-ratio (massive-activation proxy)	1.43	3.22	$\approx$ 2.3× , positive long-horizon trend
v-ratio (value-norm)	1.00	0.69	net drain, non-monotone

Warmup does not explain the rise. The h-ratio climbs from 2.40 at about 57M tokens to 3.22 at 1B, long after the 3% warmup window closes, though not monotonically at probe resolution, so we report it as a positive long-horizon trend. In Figure 1, at right, the violet trajectory climbs and never moves rightward. The v-ratio ends below its initialization value, but about 75% of that drop happens in the first 57M tokens and it then recovers part of it, so we report it as supporting context rather than ongoing emergence. **The massive-activation proxy grows about 2.3× across a full billion tokens of multimodal training while attention concentration never leaves zero (n = 1 seed, §5).**

3.3 No consistent-sign head-level relationship between concentration and value-norm

A tight coupling between concentration and value-drain should at minimum hold the sign of their per-head correlation constant across regimes. It does not. Table 3 reports the Pearson r between $\text{attn} \rightarrow \text{pos0}$ and value-norm ratio at each arm’s final checkpoint, over the 90 (layer, KV-group) observations: a value vector is shared by 3 query heads, so we average a group’s query-head attention rather than triplicate its value observation (§2). Appendix Fig. A2 plots the scatter.

Table 3 — $r(\text{attn} \rightarrow \text{pos0}, \text{v-ratio})$, final checkpoint. Descriptive only, no p-values.

baseline	g1gate	sigmoid	textinit	RF	pooled
+0.76	+0.57	−0.04	−0.79	+0.51	−0.20

These correlations are **descriptive statistics, not inferential ones**, and we deliberately report no significance tests: heads within a layer are not independent, and each arm rests on a single seed here, so a p-value would mislead. Collapsing to the 90 KV groups removes the pseudoreplication a per-query-head reading would introduce and leaves the picture unchanged (Appendix D).

The pattern is about **sign**, which pseudoreplication does not manufacture. Heads that attend more to position 0 have *larger* value norms in the baseline regime and *smaller* ones under text initialization, and the pooled correlation is weak only because arms of opposite sign cancel. We therefore claim no consistent-sign relationship across arms. We do not claim the absence of a coupling law: a fixed coupling would show one sign everywhere, and it does not.

3.4 Ordering: norm signatures lead, and concentration is late, mirrored, or absent

Sink birth & ordering: norm signatures lead; concentration is late, mirror-imaged, or never

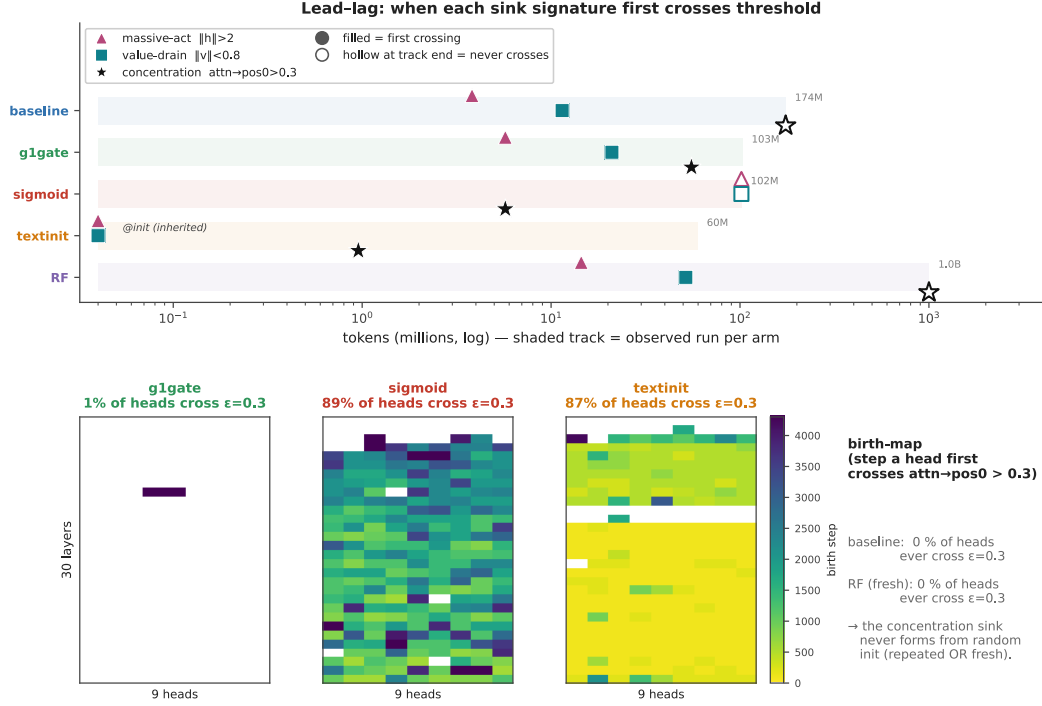


Figure 2: Sink birth and ordering. Seed 0 throughout. *Top*: time-to-event tracks spanning the tokens over which we observed each arm (60M to 1B). A filled marker is the first probe at which a signature crossed its per-head threshold ($h > 2$, $v < 0.8$, $\text{attn} \rightarrow \text{pos0} > 0.3$). A hollow marker at a track's end means it never crossed. **Crossing times are interval-censored** at the 100-step probe cadence, so two signatures crossing within one interval should not be read as ordered. *Bottom*: birth-maps, the step at which each (layer, query-head) pair first crosses the concentration threshold. No baseline and no RF head ever crosses. 1% of g1gate heads do, against 89% for sigmoid and 87% for textinit.

Dense probing lets us timestamp each signature's arrival (Figure 2), and the ordering stays consistent inside each arm family. In the softmax arms with randomly initialized decoders, *baseline* and *g1gate* and *RF*, the norm signatures come early: the residual-norm ratio crosses within the first few to about 15M tokens and value-drain follows within about 50M. Concentration never comes. No baseline or RF head ever crosses it, and g1gate reaches 1% of heads, a single-layer blip late in training. *sigmoid* mirrors that pattern, with concentration crossing at about 6M tokens and 89% of heads eventually, while neither norm signature ever crosses. *textinit* inherits its norm signatures rather than growing them: value-drain and an elevated residual-norm ratio are both present at 0 tokens, imported with the pretrained text-LM weights. Concentration is *not* inherited the same way, since Sink^{0.3}₁ is 0.000 at step 0 and crosses before 1M tokens. Even in the arm that imports the most sink structure, the norm signatures precede concentration.

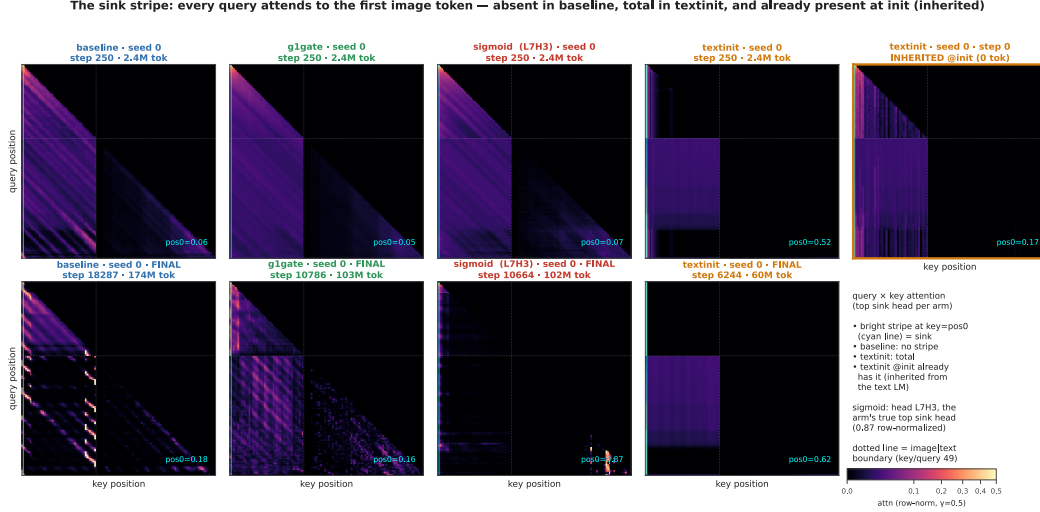


Figure 3: The sink stripe. Query $\times$ key attention of each arm's top sink head, row-normalized, seed 0. Panels carry their seed, step and token count. Top row is early training (step 250, about 2.4M tokens), bottom row the final checkpoint. The cyan line marks key = pos0, the dotted line the image-to-text boundary. The pos0 stripe is *absent* in baseline throughout and strong in textinit (attn $\rightarrow$ pos0 = 0.62 at its final checkpoint). The rightmost panel shows it *already present at step 0* in textinit, inherited from the text language model. The *sigmoid* column uses head **L7H3**, that arm's top sink head on the fixed probe batch. Its heat maps come from an auxiliary streaming batch, while every printed sigmoid number comes from the fixed probe batch (Appendix D).

Figure 3 shows the same result at the level of a single head. The vertical stripe at key = pos0 marks every query attending to the first image token. It is absent in *baseline* at both checkpoints and strong in *textinit*, and the rightmost panel carries the most weight: **the stripe is already there at step 0 in textinit**, imported with the text-LM weights before the model has seen one image.

The *sigmoid* arm needs one measurement note, because row-normalization changes the object being measured and Gu et al. [6] state their result for *unnormalized* sigmoid attention. The concentration this arm develops is a **relative** reallocation of a shrinking gate budget onto position 0, not the growth of a large absolute mass there (Appendix D). We report the row-normalized view in the tables so the arms stay comparable.

Attention-entropy collapse, the text-LM literature's usual correlate of concentration, separates the same way: only *sigmoid* and *textinit* collapse (Appendix Fig. A3). It tracks the concentration axis, not the norm axes.

The signatures also dissociate in position. A per-position scan across all three *textinit* seeds shows they need not share a token. At seed 0 they coincide at position 0. At seed 1 the attention maximum stays there while the residual peak moves to pos1 and the value minimum to pos5. At seed 2 they separate furthest, attention at **pos1** and both norm extrema at **pos13**. The arms with randomly initialized decoders behave differently: position 0 stays the maximum-mass token in *baseline*, *glgate* and *sigmoid* at every seed scanned, and in *RF* the attention argmax sits at pos1 with mass 0.100 against 0.083 at position 0, a diffuse profile rather than a sink (Appendix C).

We report this as a supporting observation, not a second headline. It has one consequence for measurement: because all three metrics anchor on position 0, the *textinit* magnitudes at seeds 1 and 2 are read at a token that is no longer the peak and therefore **understate** the arm's true peaks, a further reason to report that arm as a range and a median (§3.1, §5). The anchoring holds for the randomly initialized decoders, which carry the central negative result. Appendix E sets out, as untested hypotheses, how each lever might move a different axis.

4. Related Work

Attention sinks and their companions in text language models. Gu et al. [6] give the canonical account: the sink token acts “more like key biases, storing extra attention scores, which could be non-informative and not contribute to the value computation.” They report small key and value norms at the sink as part of the same phenomenon, connect it to massive residual-stream activations [2, 5], and show that unnormalized sigmoid attention prevents sink formation in text models up to 1B parameters. Our *sigmoid* arm builds on that result. Guo et al. [4] call the concentration and value-drain coupling “active-dormant” heads. Queipo-de-Llano, Arroyo et al. [7] make the strongest unity claim and the only genuinely *causal* one: massive activations mathematically require representational compression, and ablating a model’s layer-0 massive activation removes both compression valleys and sink formation. Across Pythia checkpoints the three appear together near step 1000 and stay synchronized. We reserve “causally unified” for that result alone. Peng et al. [24] trace a first-position sink circuit emerging early in from-scratch text pretraining, again in text, without separating the signatures.

Prior decoupling results: text-only, two axes. Two papers already separate pairs of these signatures, and we scope our claim around them. Sun, Canziani, LeCun and Zhu [9] show that massive activations and attention sinks are dissociable architectural artifacts: a change of normalization scheme crushes the massive-activation spike while the sink ratio survives. Chen and Yao [10] decouple the same pair from the opposite direction and from scratch, where a value-scale intervention in 0.1–0.3B text models keeps the sinks and suppresses massive activations. Neither treats value-norm drain as a third axis, and neither is multimodal. Fesser et al. [23] give the closest external support for tracking the value norm separately: one sink pattern can hide two algorithms, an “adaptive nop” (the head suppresses its own update by routing to a null token) and a “broadcast” that redistributes global information, and nop sinks show negligible value norms where broadcast sinks produce low-rank outputs. That work diagnoses trained vision transformers rather than tracking emergence, but it independently motivates treating the value norm as diagnostic. Qiu et al. [8] argue the opposite case, that outlier-driven rescaling by attention and residual sinks is essential to stable training. The field has settled neither question.

Multimodal sinks: mostly frozen backbones, inference time. Luo et al. [11] identify high-norm attention-sink tokens originating in the vision transformer and separate ViT-propagated from LLM-emerged sinks. Their Appendix A.4 tracks sink-dimension magnitudes across alignment checkpoints, so a training-time view of multimodal sinks has precedent, using a frozen vision transformer and a pretrained language model, and following one magnitude rather than the three signatures. Choi et al. [12] likewise distinguish vision-sinks from language-sinks in a frozen model and gate them by layer. Both establish that multimodal sinks have distinct vision-side and language-side origins, which our *textinit* inheritance result fits. What is missing there is dense, joint tracking of concentration, value-norm and residual-norm as separate quantities in a decoder that starts from random weights. Vision transformers also grow high-norm “register” tokens of their own [25], which is why the pretrained encoder gets its own limitation in §5. A separate line ties these signatures to hallucination and grounding failure in deployed models and intervenes at inference time [13–16]. That line is the practical reason it matters which signature a mitigation moves.

The gating lever. Qiu et al. [20] introduce the head-specific elementwise sigmoid gate on attention output that our *gate* arm adapts. In text models it “largely reduces the attention score allocated to the first token and decreases massive activations” while improving quality. Our variant differs in one way that matters: they use ordinary initialization, we initialize at exactly zero, so the gate opens at $\sigma(0) = 0.5$ and applies a half-scale factor at step 0 (§2), leaving gating and initial scaling confounded. With that caveat, concentration is already absent in our baseline, so the axis on which the gated arm differs is the value-norm axis. The drain becomes milder, from 0.69–0.72 to 0.81–0.85, rather than disappearing. That stays invisible unless the three signatures are logged separately.

Positioning. The closest prior work separates at most two of the three axes, in text-only models, and the multimodal studies work mostly on frozen ones. Our claim is the conjunction: **dense, joint**

tracking of concentration, value-norm drain, and the residual-norm ratio as separately measured quantities, in multimodal pretraining with a randomly initialized decoder, which adds value-norm drain as a third axis beyond the massive-activation-vs-sink dissociations shown in text models [9, 10]. Decoupling itself is not ours, and prior multimodal work could study emergence if it chose to.

5. Limitations

Metrics anchored on position 0. We measure all three signatures at the first image token, checked by per-position attention *mass* rather than *argmax* alone. Position 0 is the maximum-mass token in every arm at seed 0, and stays so for *baseline*, *glgate* and *sigmoid* at every seed we scanned. It does **not** in *textinit*, where at seed 1 the residual peak moves to pos1 and the value minimum to pos5, and at seed 2 the attention maximum sits at pos1 while both norm extrema sit at pos13 (Appendix C). Our pos0-anchored magnitudes for *textinit* at those seeds therefore *understate* its peaks, one more reason to report that arm as a range and a median and to treat its corner rather than its magnitudes as the claim. In *RF* the attention *argmax* sits at pos1, but with mass 0.100 against 0.083 at pos0, a diffuse profile rather than a displaced sink, so the RF result does not depend on the anchor.

The gate arm carries a scale confound. We initialize the G1 gate at exactly zero, so it opens at $\sigma(0) = 0.5$ and halves attention output at step 0, where Qiu et al. [20] use ordinary initialization (§2). Gating and initial output scaling are confounded in our *glgate* arm, and its differences from baseline cannot be attributed to gating alone. A scale-matched control is future work.

Pretrained vision encoder: no arm is fully from scratch. The SigLIP encoder is pretrained and trainable in every arm, and *textinit* also uses a pretrained decoder. What we study is vision–language pretraining with randomly initialized decoders, not from-scratch training of the whole model, and we word it that way throughout. Vision transformers grow high-norm register tokens of their own [25], and sinks can propagate from a vision transformer into a large vision–language model [11], so part of our residual-norm signal could be inherited rather than decoder-formed. Our defense is the trajectory. The h-ratio starts at 1.0–1.4 and *rises* to 3.22 across 1B tokens in RF, where inheritance from a static encoder predicts a high, flat h-ratio from step 0. A randomly initialized vision transformer would isolate the decoder entirely, and is future work.

The h-ratio is a proxy, not a measurement of massive activations. Massive activations are normally defined by channel-level outliers [2, 5]. We measured a position-specific residual-norm ratio and never computed channel-level statistics, so we report the h-ratio as a massive-activation proxy (§2). A large h-ratio is consistent with massive activations without establishing them.

Token scale. Our runs reach at most 1B tokens per arm, against the roughly 5B canonical in the text-LM sink literature [6]. Sink emergence is early relative to that budget, and text-LM sinks form near step 1000 [7], far inside our range. We still cannot rule out that a signature absent at 1B emerges later.

Reproducibility of *textinit* magnitudes. The massive-activation proxy of *textinit* is seed-sensitive, spanning an h-ratio of 5.5–42.5 across three seeds, with seed 0 the outlier on every signature. The corner is the reproducible claim: strong concentration, strong drain, large residual-norm ratio. No specific magnitude is.

Provenance and seed count. We take the seed-0 raw probes for the four-arm comparison from a checksummed archive summary rather than re-derive them first-hand. We *did* re-derive seeds 1 and 2 independently, and that audit caught a metric-labeling error in an earlier internal consolidation (Appendix G). *baseline* and *sigmoid* have two seeds, *glgate* and *textinit* three, RF a **single seed**. RF also contains one **weights-only optimizer restart at about 57M tokens** that an out-of-memory error forced: the weights were reloaded and the AdamW moment estimates discarded, so RF is not one uninterrupted optimizer trajectory. The audit verified continuity across that seam, where the v-ratio and h-ratio are identical at the shared checkpoint, a double-covered 600-step overlap diverges only within probe noise, and concentration reads 0.000 on both sides.

Concentration was reproducibly zero across both seeds of the repeated-data baseline, which we take as adequate support for the negative claim. A second fresh-data seed would strengthen it.

What cannot be checked on RF. RF has no distinct seen split, so we cannot run for it the `val_seen / val_unseen` comparison that exposes memorization in the repeated arms. The weaker statement its data supports is that its held-out loss falls throughout and never turns upward (§2). Its streaming shuffle buffer also dropped from 1500 to 500 examples partway through, again for memory reasons, so stream ordering is not homogeneous across the run, though the $\text{Sink}^{0.3}_1 = 0.000$ result and the h-ratio rise hold within each regime separately. The RF probe batch is still the **fixed repeated-the_cauldron tail** used by the other arms. That keeps RF’s signatures comparable to theirs, which is what the negative result needs, at the cost of measuring them on data from the other distribution.

Domain shift in the fresh-data control. RF reduces repetition by changing dataset, which introduces a domain-shift confound in its place. We took that trade deliberately, because repetition is the dominant confound, and chose the fresh pool to minimize shift. The under-3% overlap figure is an estimate from config-level composition, not image-level deduplication (§2), and we did not run a third, domain-matched control. That control is the known follow-up.

We report no benchmark accuracy. We measure sink signatures with the probe of §2. We did not run downstream benchmark evaluation, such as MMStar, on any arm, and we make no claim about how signature dissociation relates to downstream capability.

6. Conclusion

We tracked concentration, value-norm drain, and a residual-norm ratio we read as a massive-activation proxy as three separate quantities, across multimodal pretraining with a randomly initialized decoder. They came apart everywhere we looked. Four levers produced four different signature corners. The value-norm axis alone moved in three directions. On a low-repetition run over a billion fresh tokens the proxy grew about $2.3\times$ while concentration never left zero, and the per-head relationship between the first two flipped sign across arms. The signatures also arrived in different orders, and under text initialization they sat on different tokens.

Text-LM work documents real interactions among these signatures, including a causal route from massive activations to sinks and compression valleys [7]. What our results add is that the coupling is optional. Each axis moved on its own under ordinary levers, which extends the two-way text-only dissociations [9, 10] to a third axis and a new setting. The practical consequence for interpretability and mitigation work is blunt. One signature is not a proxy for the others. A model with no attention sink can still carry a growing residual-norm asymmetry, and a gate that changes value-drain can leave the other two axes untouched.

Next steps. A randomly initialized vision encoder, to isolate what the decoder contributes to the residual-norm signal. A fresh-data run past 1B tokens, to match text-LM budgets. A scale-matched gate control, to separate gating from the half-scale confound of §2.

Reproducibility. A self-validating probe (§2) computes all signatures on a fixed probe batch, from logs taken every 100 steps, and the Appendix holds the per-seed tables. We will release the code, the probe, the run configurations, the per-run logs, and the training checkpoints on acceptance. We withhold the links for double-blind review.